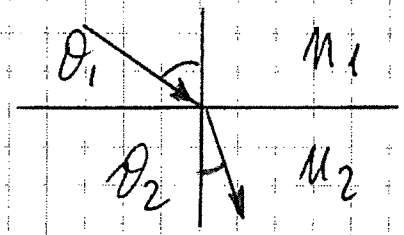
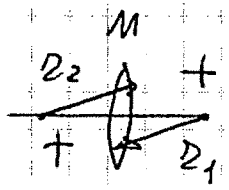
Geometrical optics

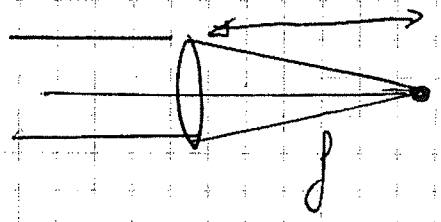


$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Lensmaker's eq.

$$\frac{1}{f} = (n-1)\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$$





Lens law

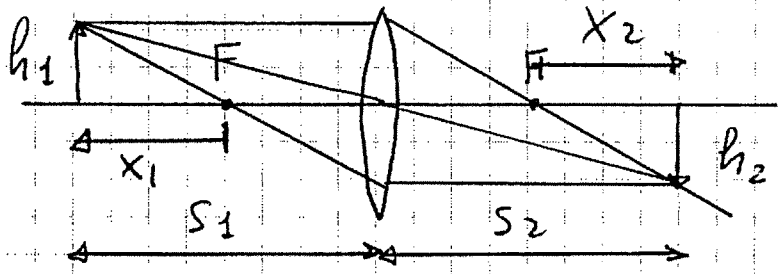
$$1/s_1 + 1/s_2 = 1/f$$

$$M_L = -\frac{|h_2|}{|h_1|} = -\frac{|s_2|}{|s_1|}$$

Newton's equation

$$x_1 x_2 = -f^2 \quad \begin{pmatrix} x_1 < 0 \\ x_2 > 0 \end{pmatrix}$$

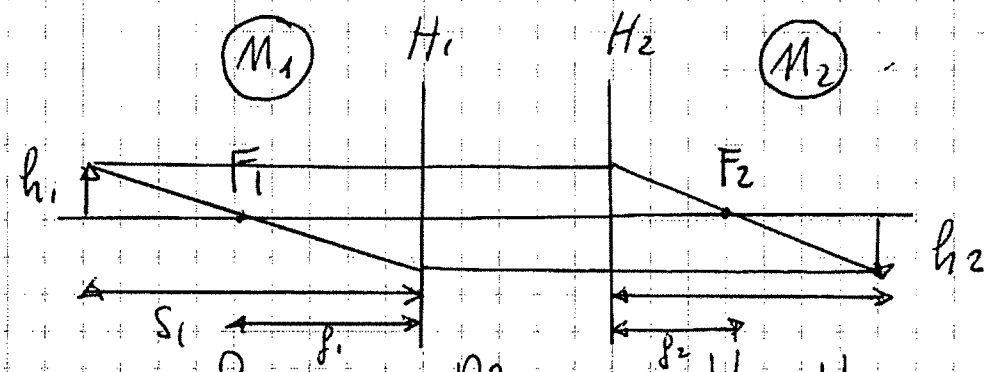
$$M_L = f/x_1 = -\frac{x_2}{f}$$



Generic optical system

$$\frac{n_1}{s_1} + \frac{n_2}{s_2} = \frac{n_1}{f_1} = \frac{n_2}{f_2}$$

$$[n_1 = n_2] \rightarrow 1/s_1 + 1/s_2 = 1/f$$



Princ. Planes H_1, H_2

Lateral magnification $H_1 \leftrightarrow H_2$ $M_L = +1$

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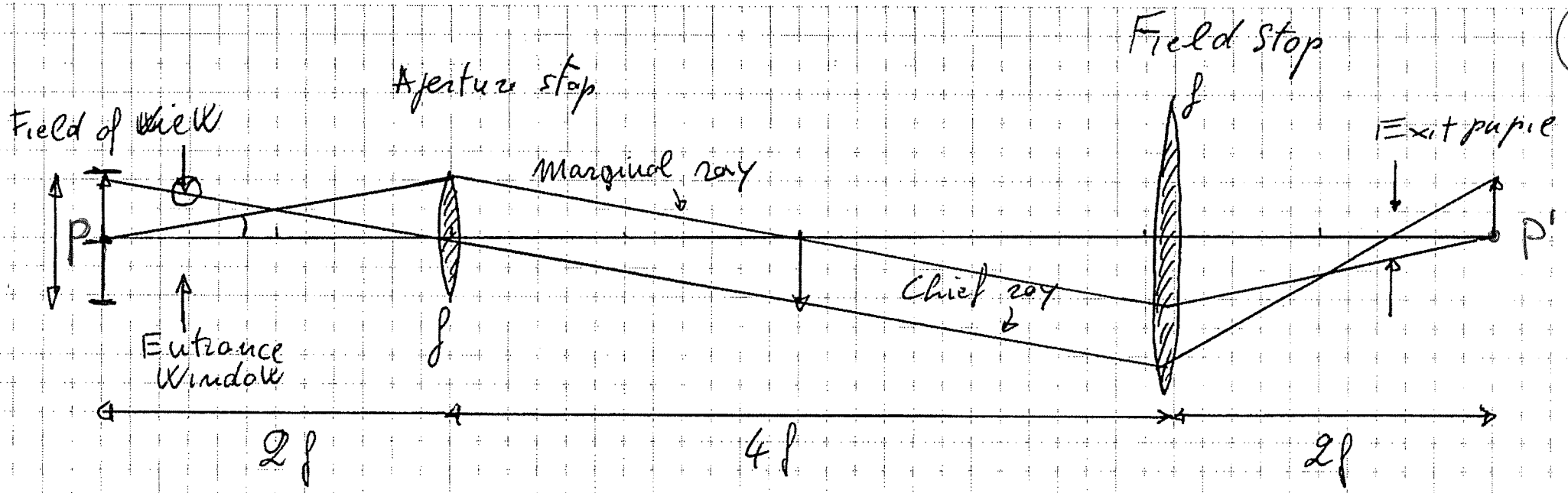

- Aperture stop: limits the amount of light collected by the opt. system
- Field Stop: limits the field of view of the optical system
- Entrance pupil: Image of the aperture stop in the object space
- Exit pupil: Image of the aperture stop in the image space
- Entrance Window: Image of the field stop in the object space
- Exit Window: Image of the field stop in the image space

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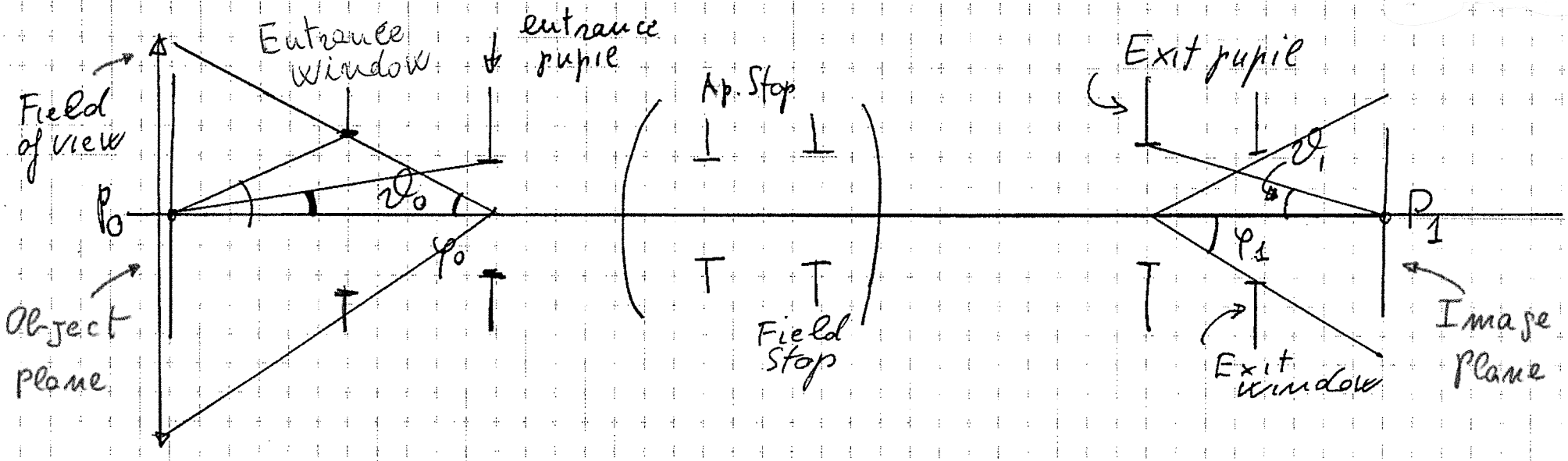
General method to find stops, pupils and windows



- The entrance pupil is the image of that physical stop that subtends the smallest angle at an axial point in the object plane (P_0)
- The entrance window is the image of that physical stop that subtends the smallest angle at the centre of the entrance pupil
- $2\vartheta_0$ is the angular aperture of the optical system
- $2\varphi_0$ is the angular field of view of the optical system

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